CLASSIFICATION ASSIGNMENT

Chronic Kidney Disease Prediction

Objective

Build and compare classification models that predict whether a patient has chronic kidney disease.

Dataset Size	Input Features	Best Result Shown
399 rows	**25 columns**	**Random Forest**

Project Overview

This assignment focuses on predicting chronic kidney disease using supervised machine-learning classification. The dataset contains 399 records and 25 columns.

Data Preparation and Modelling Approach

- Prepared the dataset for modelling by converting string-based or categorical values into numerical form.
- Applied standard scaling to place numerical features on a consistent scale before model training.
- Used grid search to evaluate candidate hyperparameter combinations and identify strong configurations.
- Compared Random Forest, Support Vector Machine (SVM), Logistic Regression, and Decision Tree classifiers.

Results Summary

Model	Accuracy	ROC–AUC	Observation
Random Forest	0.99	1.0000	Strongest result shown in the assignment
SVM	0.97	0.996	Balanced performance across both classes
Logistic Regression	0.98	0.9978	Very strong linear-model baseline
Decision Tree	0.95	0.9425	Lowest result among the displayed models

Result interpretation: Based on the displayed test metrics, Random Forest produced the strongest overall result, with 0.99 accuracy and a ROC–AUC score of 1.0.

Model Evaluation Results

1. Random Forest

Grid search produced the strongest displayed result. The classification report shows 0.99 accuracy, while the ROC–AUC score is 1.0.

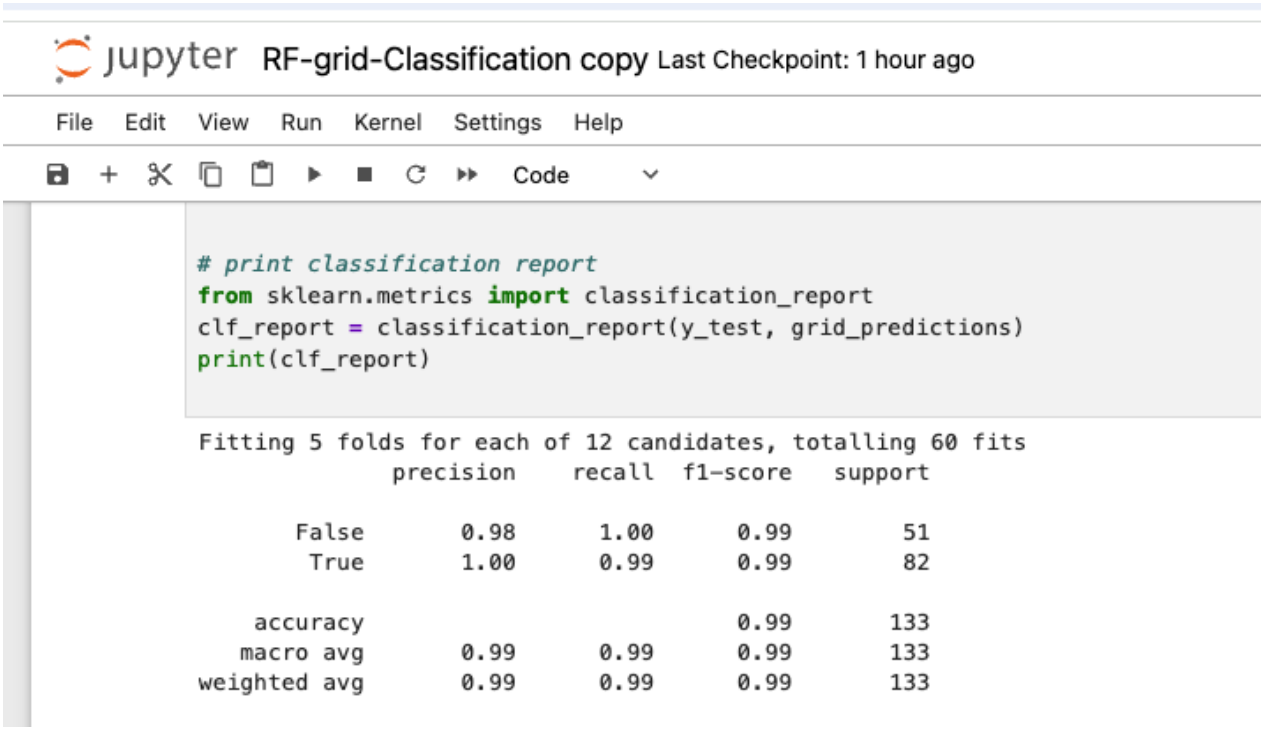


Figure 1. Random Forest classification report

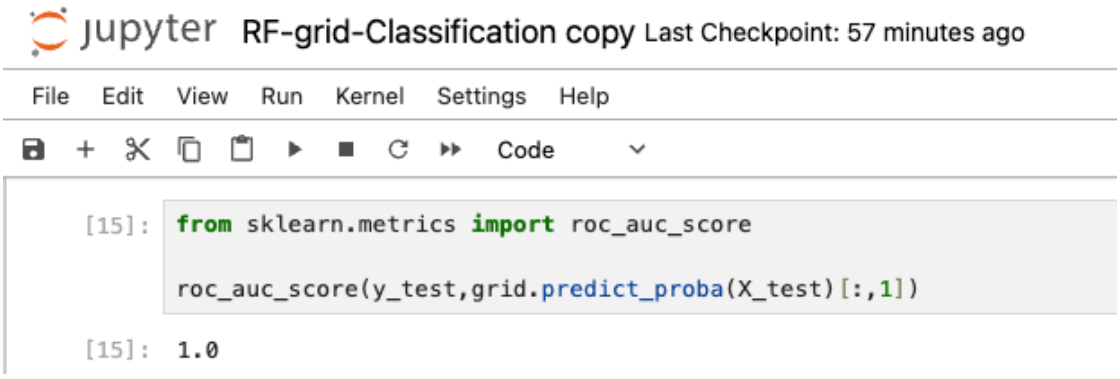


Figure 2. Random Forest ROC–AUC score

Additional Model Results

2. Support Vector Machine (SVM)

The SVM model achieved 0.97 accuracy in the displayed classification report.

jupyter SVM-kernel Classification

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Code

```
re=grid.cv_results_
#print(re)
grid_predictions = grid.predict(X_test)


from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, grid_predictions)



# print classification report
from sklearn.metrics import classification_report
clf_report = classification_report(y_test, grid_predictions)
print(clf_report)
```

```
Fitting 5 folds for each of 40 candidates, totalling 200 fits
              precision    recall  f1-score   support

       False       0.94      0.98      0.96        51
        True       0.99      0.96      0.98        82

    accuracy                           0.97       133
   macro avg       0.97      0.97      0.97       133
weighted avg       0.97      0.97      0.97       133
```

[36]:
```
print(cm)
```

```
[[49  2]
 [ 3 79]]
```

[37]:
```
from sklearn.metrics import roc_auc_score

roc_auc_score(y_test,grid.predict_proba(X_test)[:,1])
```

[37]:
```
0.9966523194643712
```

Figure 3. SVM classification report

3. Logistic Regression

Logistic Regression achieved 0.98 accuracy and a ROC–AUC score of approximately 0.9978.

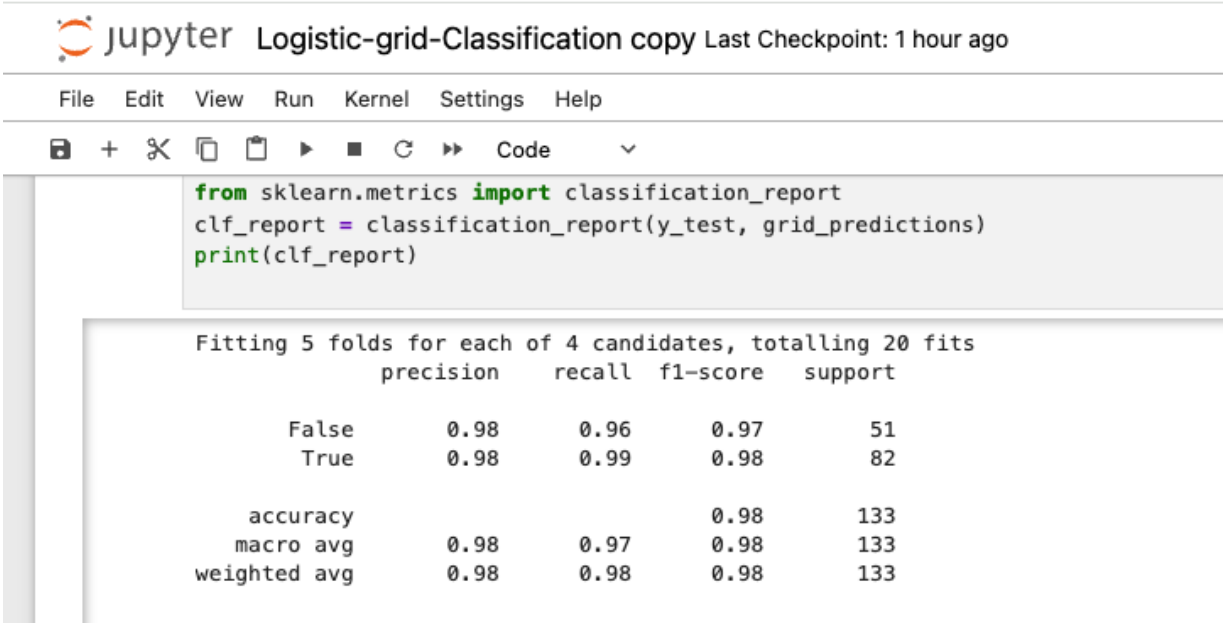


Figure 4. Logistic Regression classification report



Figure 5. Logistic Regression ROC–AUC score

4. Decision Tree

The Decision Tree model achieved 0.95 accuracy and a ROC–AUC score of approximately 0.9425.

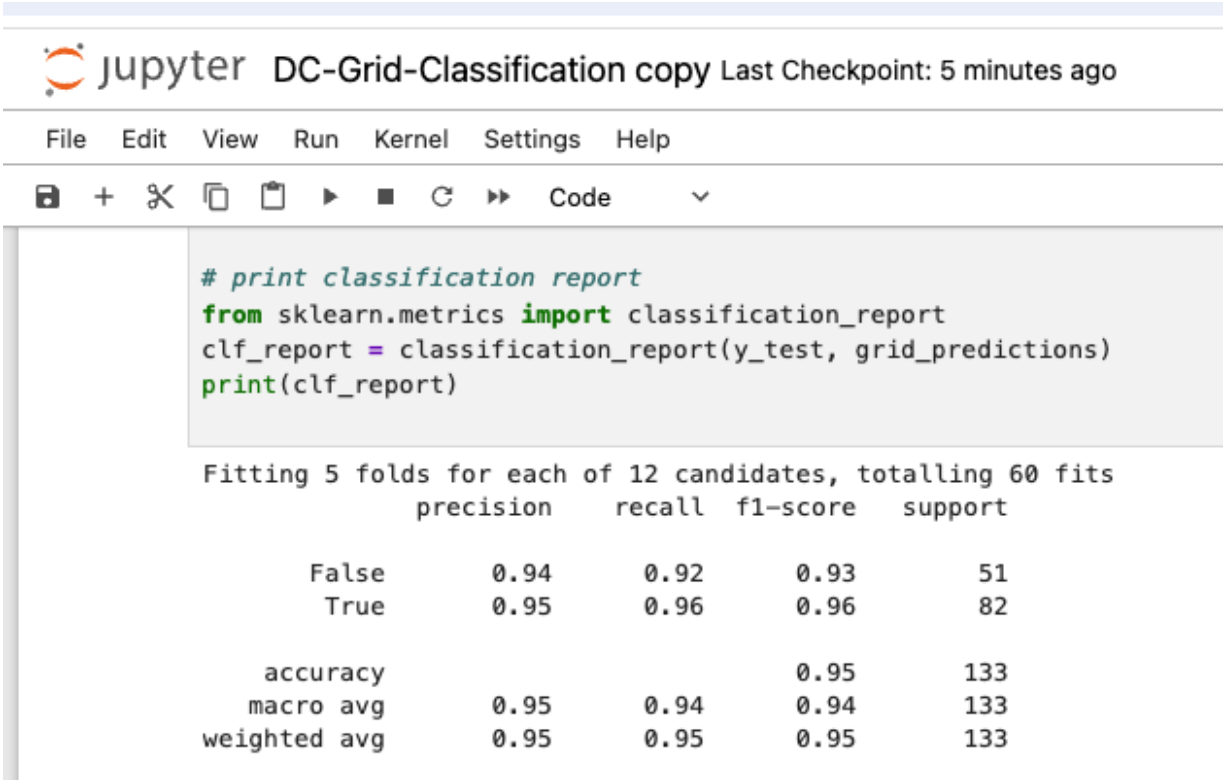


Figure 6. Decision Tree classification report

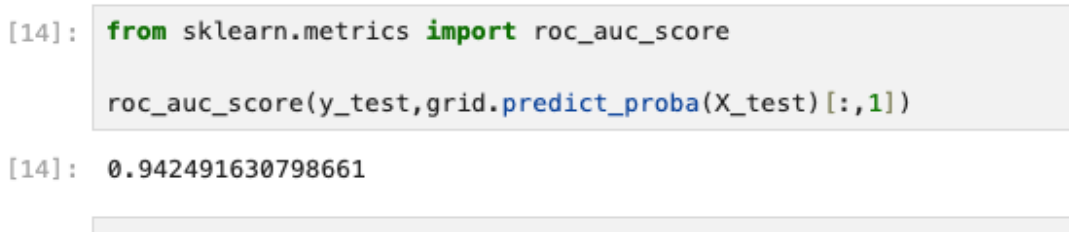


Figure 7. Decision Tree ROC–AUC score

Conclusion

Among the evaluated classifiers, Random Forest delivered the best displayed performance. Logistic Regression also performed very strongly, followed by SVM and Decision Tree. The comparison demonstrates the value of preprocessing, hyperparameter tuning, and metric-based model evaluation when selecting a classifier for chronic kidney disease prediction.